AN INTERNSHIP PROJECT REPORT

ON

SMARTBRANCH 360

Secure Branch Office Network Simulation with Python-Assisted Network Assurance

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CANDIDATE'S DECLARATION

I, Deepshikha Rani, hereby declare that the internship project report entitled “SmartBranch 360: Secure Branch Office Network Simulation with Python-Assisted Network Assurance” is an original record of the work completed by me during the Cisco Virtual Internship 2026. The project has been developed to fulfil the academic requirements of the Bachelor of Technology programme in Computer Science and Engineering at IILM University.

The network topology, Cisco IOS configurations, testing records, fault-injection exercises, Python assurance tool, and supporting documentation presented in this report were prepared for this project. Wherever external software platforms, technologies, or documentation were used, they have been identified appropriately. This report has not been submitted to any other institution for the award of a degree or diploma.

I accept responsibility for the accuracy of the technical work and for maintaining academic integrity in the preparation of this submission.

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I also acknowledge the guidance and instructional support received throughout the Cisco Virtual Internship 2026. The practical activities on switching, routing, network services, security, troubleshooting, and automation helped me translate theoretical concepts into a complete branch-office implementation. I am grateful to the faculty members and instructors who supported the learning process and encouraged systematic documentation of the work.

Finally, I thank my family and well-wishers for their continuous motivation and support during the completion of the project and preparation of this report.

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ABSTRACT

SmartBranch 360 is a secure branch-office network simulation developed in Cisco Packet Tracer and supported by a lightweight Python network assurance tool. The project addresses the limitations of a flat local-area network by separating employee, guest, server, and management traffic into VLANs 10, 20, 30, and 99. A Cisco 2911 router performs inter-VLAN routing through IEEE 802.1Q subinterfaces and provides DHCP, NAT/PAT, access-control enforcement, and SSH management. Two Cisco 2960 switches extend the segmented design, while an access point provides wireless connectivity for guest clients.

The security architecture permits normal employee access to internal and simulated external services, blocks guest traffic from protected internal networks, and restricts remote device administration to the management workstation at 10.10.99.10. The implementation is validated through positive and negative connectivity tests, Cisco IOS show-command evidence, DHCP and DNS checks, NAT translation inspection, and management-only SSH verification.

To demonstrate operational troubleshooting, five controlled faults were injected: an incorrect employee gateway, a missing Guest VLAN on a trunk, an incorrect Guest DHCP gateway, an ACL rule blocking DNS, and a missing NAT outside role. Each fault was diagnosed from observable symptoms and command evidence before the correct configuration was restored. The Python checker reads the intended state from requirements.yaml, parses captured Cisco outputs, and reports PASS, WARN, or FAIL findings. The healthy data set produces 25 passes with no warnings or failures, while the F02 sample correctly detects VLAN 20 missing from SW1 Gi0/2. The project therefore demonstrates not only a functional network, but also a documented, testable, reproducible, and diagnosable implementation.

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1. INTRODUCTION

Modern branch offices require reliable user connectivity without treating every device as equally trusted. SmartBranch 360 was developed as a complete small-branch network in Cisco Packet Tracer to demonstrate how segmentation, centralized routing, controlled services, security policy, troubleshooting, and automation can be combined in one practical implementation.

The project uses four functional networks: Employee, Guest, Server, and Management. These networks are implemented as separate VLANs and IPv4 subnets, with routing centralized on R1-BRANCH through a router-on-a-stick design. DHCP and DNS support client operation, NAT/PAT represents access to a simulated external network, an inbound ACL isolates guest users from internal resources, and SSH is limited to an authorized management host.

A major emphasis of the work is proof. Healthy operation is supported by endpoint tests and Cisco IOS command captures, while five controlled faults demonstrate a repeatable diagnose-repair-retest process. A Python assurance utility then compares captured device state against machine-readable YAML requirements. This summary intentionally includes only representative evidence; the complete screenshot set remains available in the project repository.

Project repository: deeps2710/SmartBranch360

2. PROBLEM STATEMENT

A branch office must support employees, visitors, shared services, and administrators, but a single flat LAN cannot enforce appropriate trust boundaries. In such a network, guest devices may reach internal hosts, broadcast traffic affects every endpoint, and troubleshooting becomes difficult because addressing and security responsibilities are not clearly separated.

The network must therefore provide role-based segmentation while preserving the services each role requires. Employees need internal and external access, guests need wireless access to the simulated Internet without access to protected networks, servers require stable addressing, and network devices must be manageable only from the designated administration workstation.

Operational reliability is a second concern. Small configuration mistakes can create selective failures that are not obvious from the topology alone. A wrong gateway can isolate one client; a missing allowed VLAN can break a complete segment; an incorrect DHCP option can affect every renewed lease; an ACL can block a single application service; and an incomplete NAT role can prevent translations. SmartBranch 360 addresses both the design problem and the need for evidence-based diagnosis.

Required outcome

- A segmented branch topology with predictable addressing and routing.
- Explicit guest-isolation and management-access policies.
- Verified client services and controlled external connectivity.
- Repeatable fault-remediation evidence and automated configuration assurance.

3. OBJECTIVES

1. Design a clearly labelled branch-office topology using one router, two switches, a wireless access point, servers, and representative client devices.
2. Create separate Employee, Guest, Server, and Management broadcast domains using VLANs 10, 20, 30, and 99.
3. Assign one /24 IPv4 subnet and a predictable default gateway to each internal VLAN.
4. Implement inter-VLAN routing with IEEE 802.1Q router subinterfaces.
5. Provide DHCP, DNS, guest wireless connectivity, and NAT/PAT toward a simulated external network.
6. Deny Guest access to Employee, Server, and Management networks while permitting required external traffic.
7. Restrict SSH management of network devices to ADMIN-PC at 10.10.99.10.
8. Verify healthy operation through positive and negative endpoint tests and Cisco IOS show commands.
9. Inject and repair five representative configuration faults using a controlled troubleshooting process.
10. Validate the captured device state against requirements.yaml through a Python and PyYAML assurance tool.
11. Maintain a reproducible project history and evidence set through Git and GitHub.

4. TECHNOLOGIES USED

The project combines a network simulator, vendor command-line configuration, a small automation utility, a machine-readable requirements file, and version-control tooling. Each technology has a distinct role in the evidence chain.

Technology	Role in SmartBranch 360
Cisco Packet Tracer	Models the complete branch topology, endpoint behaviour, wireless access, and network services in a controlled simulation.
Cisco IOS CLI	Configures VLANs, trunks, subinterfaces, DHCP, NAT/PAT, ACLs, and SSH; also supplies operational show-command evidence.
Python	Implements the network assurance checker and converts captured IOS output into structured PASS, WARN, and FAIL findings.
PyYAML	Loads requirements.yaml so the checker validates observed state against an explicit intended design.
Git	Records local project revisions and maintains a traceable history of configuration evidence, code, and documentation.
GitHub	Hosts the submission repository and makes the topology, checker, reports, and evidence accessible from one location.

Primary implementation environment: Cisco Packet Tracer with Cisco 2911 and 2960-24TT device models.

5. PROPOSED NETWORK ARCHITECTURE

The proposed architecture follows a compact hierarchical branch design. R1-BRANCH is the Layer-3 control point and external edge. SW1 acts as the primary access and aggregation switch, while SW2 extends the required VLANs to additional employee devices and the Guest wireless access point. The internal server and the dedicated administration host remain on protected segments.

Architecture principles

- Segmentation by role: user, guest, service, and management traffic occupy separate Layer-2 broadcast domains.
- Centralized routing: all internal VLAN gateways terminate on R1-BRANCH, creating one auditable policy point.
- Controlled trust: Guest traffic is evaluated inbound before it can enter protected internal networks.
- Stable infrastructure addressing: router gateways, switch management interfaces, servers, and ADMIN-PC use reserved addresses.
- Testable intent: VLANs, gateways, trunks, NAT roles, DHCP gateways, ACL placement, and SSH restrictions are represented in requirements.yaml.

Traffic policy

Source	Destination	Policy	Purpose
Employee	SRV1 and external network	Allow	Supports normal branch work and service access.
Guest	Employee VLAN 10	Deny	Protects employee endpoints from visitor traffic.
Guest	Server VLAN 30	Deny	Protects the internal service segment.
Guest	Management VLAN 99	Deny	Prevents access to network administration resources.
Guest	External network	Allow	Provides the intended guest Internet experience.
ADMIN-PC	R1, SW1, and SW2	Allow SSH	Restricts remote administration to one authorized source.

6. DEVICE INVENTORY

The simulated branch contains fourteen active infrastructure and endpoint devices. The inventory below identifies the Packet Tracer type, quantity, addressing, and functional role of each device or endpoint group.

Device / Group	Qty.	Addressing	Role
R1-BRANCH (Cisco 2911)	1	VLAN gateways; G0/1 = 203.0.113.1/24	Inter-VLAN routing, DHCP, NAT/PAT, ACL enforcement, and SSH.
SW1 (Cisco 2960-24TT)	1	10.10.99.2/24	Primary access switch; router trunk and SW2 trunk.
SW2 (Cisco 2960-24TT)	1	10.10.99.3/24	Secondary access switch; employee and Guest AP connectivity.
SRV1 (Server-PT)	1	10.10.30.10/24	Internal service host and Employee DNS server.
INTERNET-SRV (Server-PT)	1	203.0.113.10/24	Simulated external web/DNS destination.
AP-GUEST (AccessPoint-PT)	1	Layer-2 bridge in VLAN 20	Wireless access using SSID SmartBranch-Guest.
EMP-PC1 to EMP-PC5	5	DHCP in 10.10.10.0/24	Wired Employee VLAN clients.
GUEST-LAP1 to GUEST-LAP2	2	DHCP in 10.10.20.0/24	Wireless Guest VLAN clients.
ADMIN-PC	1	10.10.99.10/24	Authorized management and SSH workstation.

7. VLAN AND IP ADDRESSING PLAN

Four internal /24 networks provide clear role separation. In every subnet, .1 is the router gateway and the first ten addresses are excluded from DHCP to protect infrastructure and reserved assignments.

VLAN	Name	Subnet	Gateway	Primary Use
10	EMPLOYEE	10.10.10.0/24	10.10.10.1	EMP-PC1 to EMP-PC5
20	GUEST	10.10.20.0/24	10.10.20.1	AP-GUEST and guest laptops
30	SERVER	10.10.30.0/24	10.10.30.1	SRV1 and protected services
99	MANAGEMENT	10.10.99.0/24	10.10.99.1	ADMIN-PC, SW1, and SW2

Infrastructure and service addresses

Device / Interface	IPv4 Address	Assignment
SW1 VLAN 99 SVI	10.10.99.2/24	Static management
SW2 VLAN 99 SVI	10.10.99.3/24	Static management
ADMIN-PC	10.10.99.10/24	Static authorized SSH source
SRV1	10.10.30.10/24	Static internal server / Employee DNS
R1-BRANCH G0/1	203.0.113.1/24	Static NAT outside interface
INTERNET-SRV	203.0.113.10/24	Static simulated external server / Guest DNS

8. PHYSICAL AND LOGICAL TOPOLOGY

Physically, R1-BRANCH connects to SW1 and to the simulated external server network. SW1 connects employee devices, SRV1, ADMIN-PC, and SW2. SW2 connects two additional employee systems and AP-GUEST, which bridges the wireless guest laptops into VLAN 20.

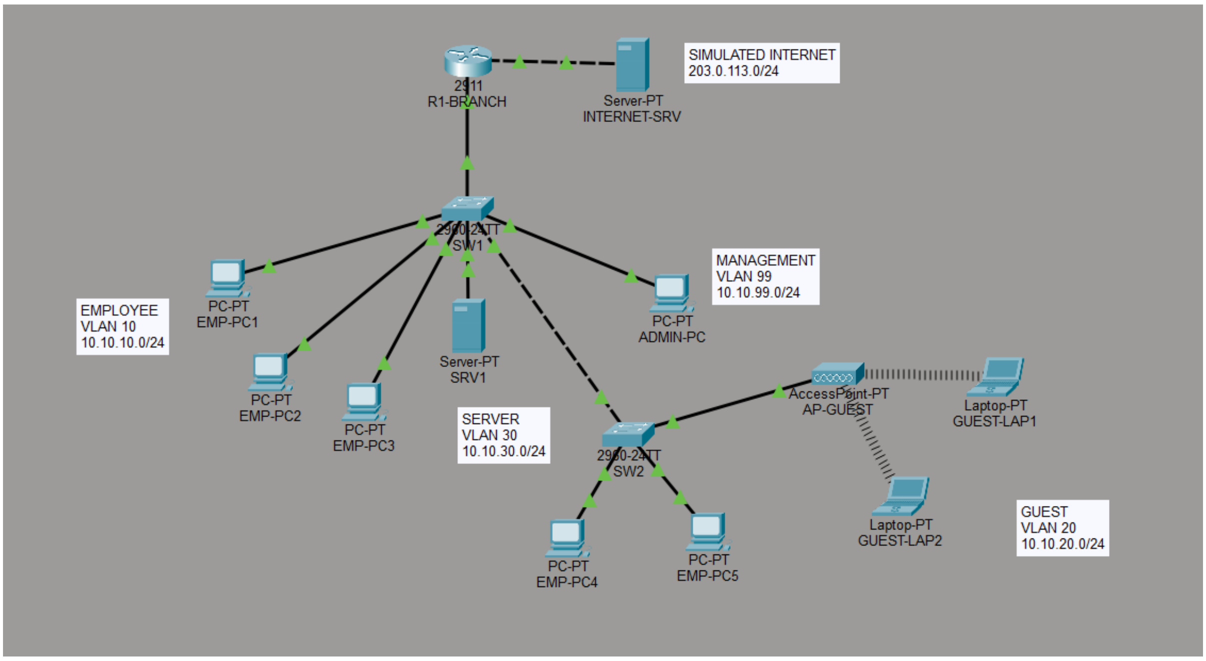


Figure 1 — Complete SmartBranch 360 topology.

Logically, the SW1-to-router and SW1-to-SW2 links carry VLANs 10, 20, 30, and 99 as 802.1Q trunks. Access links place each endpoint in one VLAN. R1-BRANCH terminates one subinterface per VLAN and routes traffic according to the Guest ACL, management access restrictions, and NAT policy.

Link	Mode	Allowed VLANs / Purpose
R1 G0/0 ↔ SW1 Gi0/1	802.1Q trunk	10, 20, 30, 99; router-on-a-stick
SW1 Gi0/2 ↔ SW2 Gi0/1	802.1Q trunk	10, 20, 30, 99; VLAN extension
R1 G0/1 ↔ INTERNET-SRV	Routed external link	203.0.113.0/24; NAT outside
SW2 access port ↔ AP-GUEST	Access	VLAN 20 Guest wireless segment

9. IMPLEMENTATION

9.1 VLAN Configuration

VLANs 10, 20, 30, and 99 were created on both switches using identical names. Employee and infrastructure ports were assigned to the appropriate access VLAN so that endpoint placement remained explicit and auditable.

```
vlan 10
 name EMPLOYEE
vlan 20
 name GUEST
vlan 30
 name SERVER
vlan 99
 name MANAGEMENT
```

9.2 Trunk Configuration

SW1 Gi0/1 carries the four VLANs to R1-BRANCH. SW1 Gi0/2 and SW2 Gi0/1 form the inter-switch trunk. The allowed list is deliberately explicit because the F02 exercise tests the effect of removing VLAN 20 from this baseline.

```
interface GigabitEthernet0/2
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,99
```

9.3 Router-on-a-Stick

R1-BRANCH divides G0/0 into four 802.1Q subinterfaces. Each subinterface uses the .1 address of its subnet and is configured as a NAT inside interface. The Guest ACL is applied inbound on G0/0.20 so unwanted traffic is filtered close to its source.

```
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 10.10.20.1 255.255.255.0
 ip nat inside
 ip access-group 100 in
```

9.4 DHCP

R1-BRANCH provides role-based DHCP pools. Addresses .1 through .10 are excluded in each internal subnet. EMPLOYEE_POOL supplies gateway 10.10.10.1 and DNS 10.10.30.10; GUEST_POOL supplies gateway 10.10.20.1 and external DNS 203.0.113.10. SERVER_POOL and MANAGEMENT_POOL are also defined, while the key infrastructure devices remain statically addressed.

```
ip dhcp pool EMPLOYEE_POOL
 network 10.10.10.0 255.255.255.0
 default-router 10.10.10.1
 dns-server 10.10.30.10
```

```
ip dhcp pool GUEST_POOL
 network 10.10.20.0 255.255.255.0
 default-router 10.10.20.1
 dns-server 203.0.113.10
```

9.5 DNS

SRV1 at 10.10.30.10 is the DNS server assigned to Employee clients. Guest clients use the simulated external DNS service at 203.0.113.10, which resolves internet.test. Separating these assignments permits Guest isolation from the internal Server VLAN while retaining name resolution for the allowed external destination.

9.6 Guest Wi-Fi

AP-GUEST is connected to an SW2 access port in VLAN 20 and advertises the SSID SmartBranch-Guest. GUEST-LAP1 and GUEST-LAP2 associate wirelessly, obtain addresses from GUEST_POOL, and enter the routed network only through the Guest subinterface and its inbound ACL.

9.7 NAT/PAT

All four internal router subinterfaces are marked ip nat inside, while G0/1 at 203.0.113.1 is marked ip nat outside. Standard ACL 1 identifies the four internal /24 networks, and overload translates concurrent flows through the G0/1 address.

```
ip nat inside source list 1 interface GigabitEthernet0/1 overload
```

9.8 ACL Security

Extended ACL 100 denies Guest traffic to the Employee, Server, and Management subnets, then permits all remaining IP traffic. Applying the ACL inbound on G0/0.20 preserves external access without exposing protected internal networks.

```
access-list 100 deny ip 10.10.20.0 0.0.0.255 10.10.10.0 0.0.0.255
access-list 100 deny ip 10.10.20.0 0.0.0.255 10.10.30.0 0.0.0.255
access-list 100 deny ip 10.10.20.0 0.0.0.255 10.10.99.0 0.0.0.255
access-list 100 permit ip any any
```

9.9 SSH Management

SSH version 2, the smartbranch.local domain, a local privileged administrator, and VTY login local are configured on the network devices. Standard ACL 10 permits only host 10.10.99.10, and the VTY access-class applies that restriction inbound.

```
ip ssh version 2
ip domain-name smartbranch.local
access-list 10 permit host 10.10.99.10
line vty 0 4
 access-class 10 in
 login local
 transport input ssh
```

10. TESTING AND VERIFICATION

Verification was organized by requirement rather than by device. Positive tests confirmed required services, while negative tests demonstrated that security controls actively blocked unauthorized traffic. Cisco IOS evidence was used to support endpoint behaviour. Of more than thirty captured screenshots, Figures 2 to 6 provide the representative evidence retained in this summary.

Test	Source / Check	Expected Result	Final Status
T01	Employee DHCP	10.10.10.0/24 lease; gateway 10.10.10.1	PASS
T02	Employee → SRV1	Reach 10.10.30.10	PASS
T03	Employee / Guest → external	Reach 203.0.113.10; create translations	PASS
T04	Guest → internal networks	Blocked from VLANs 10, 30, and 99	PASS
T05	ADMIN-PC SSH	Authorized login to R1, SW1, and SW2	PASS
T06	Employee SSH attempt	Rejected by management source restriction	PASS
T07	VLAN and trunk state	VLANs 10,20,30,99 present and allowed	PASS
T08	Python healthy validation	25 PASS, 0 WARN, 0 FAIL	PASS

Representative service and security evidence

Figure 2 — Employee DHCP lease with gateway 10.10.10.1 and DNS server 10.10.30.10.

The Employee client received 10.10.10.14/24, confirming correct pool selection and delivery of the expected gateway and internal DNS server.

```
Pinging 10.10.30.10 with 32 bytes of data:

Reply from 10.10.30.10: bytes=32 time=10ms TTL=127
Reply from 10.10.30.10: bytes=32 time<1ms TTL=127
Reply from 10.10.30.10: bytes=32 time<1ms TTL=127
Reply from 10.10.30.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms
```

Figure 3 — Employee-to-Server connectivity with four successful replies from 10.10.30.10.

The successful cross-VLAN ping verifies the Employee access port, VLAN 10 trunk path, router subinterfaces, inter-VLAN routing, and the Server VLAN path.

```
Pinging 10.10.30.10 with 32 bytes of data:

Reply from 10.10.20.1: Destination host unreachable.
Reply from 10.10.20.1: Destination host unreachable.
Reply from 10.10.20.1: Destination host unreachable.
Reply from 10.10.20.1: Destination host unreachable.

Ping statistics for 10.10.30.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure 4 — Guest-to-Server access blocked by the inbound Guest ACL.

The router gateway returns Destination host unreachable for the protected server destination. This is an expected security result rather than a connectivity defect.

```
C:\>ssh -l admin 10.10.99.1

Password:


R1-BRANCH#
```

Figure 5 — Management SSH verification from ADMIN-PC to R1-BRANCH.

ADMIN-PC at 10.10.99.10 successfully reaches the router CLI, while captured negative evidence in the repository shows that an Employee source is rejected. Equivalent successful checks were captured for SW1 and SW2.

```
R1-BRANCH>show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
tcp 203.0.113.1:1025   10.10.20.12:1025   203.0.113.10:80    203.0.113.10:80
tcp 203.0.113.1:1026   10.10.10.11:1026   203.0.113.10:80    203.0.113.10:80

R1-BRANCH>show ip nat statistics
Total translations: 2 (0 static, 2 dynamic, 2 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0.10 , GigabitEthernet0/0.20 , GigabitEthernet0/0.30 ,
GigabitEthernet0/0.99
Hits: 42  Misses: 33
Expired translations: 29
Dynamic mappings:
```

Figure 6 — NAT translation evidence for Employee and Guest sessions through 203.0.113.1.

The translation table maps inside-local addresses 10.10.20.12 and 10.10.10.11 to the G0/1 address with unique transport identifiers, demonstrating PAT for concurrent internal sessions.

11. FAULT INJECTION AND TROUBLESHOOTING

Troubleshooting followed a controlled six-step method: confirm the healthy baseline, inject one fault, reproduce the symptom, collect evidence, repair only the affected configuration, and repeat the original test. Keeping one variable at a time made every conclusion traceable to a specific cause.

Fault	Injected Error	Observed Symptom	Corrective Action	Verification
F01	Employee gateway set to 10.10.10.254	Employee loses remote VLAN and external access	Restore 10.10.10.1	Internal and external ping restored
F02	VLAN 20 removed from SW1 Gi0/2	Guest cannot reach 10.10.20.1	Re-add VLAN 20 to trunk	Trunk list and Guest path restored
F03	GUEST_POOL gateway set to 10.10.20.254	Guest receives a lease but cannot route	Restore 10.10.20.1 and renew	Gateway and external ping restored
F04	ACL denies Guest UDP/53 to 203.0.113.10	IP ping works but internet.test does not resolve	Remove unintended DNS deny	Name resolution and ping succeed
F05	Remove ip nat outside on R1 G0/1	Fresh traffic creates no proper translation	Restore role and clear translations	New translation entries appear

Detailed F02 evidence chain

```
SW1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1
Gig0/2      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      10,20,30,99
Gig0/2      10,30,99

Port        Vlans allowed and active in management domain
Gig0/1      10,20,30,99
Gig0/2      10,30,99

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      10,20,30,99
Gig0/2      10,30,99
```

Figure 7 — F02 missing VLAN fault: SW1 Gi0/2 allows 10,30,99 and omits VLAN 20.

The Layer-2 evidence localizes the fault to the inter-switch trunk. VLAN 20 remains allowed on SW1 Gi0/1 but is absent from Gi0/2, which explains why the Guest segment behind SW2 loses its path to the router.

```
SmartBranch 360 Assurance - F02

$ python python_checker/checker.py --sw1-trunk sample_inputs/sw1_trunk_fault.txt

[FAIL] VLAN 20 missing from trunk SW1 Gi0/2
       Expected: 10,20,30,99
       Observed: 10,30,99
       Symptom: Guest connectivity may fail.
       Suggested fix: Add VLAN 20 to the allowed trunk VLAN list.

Checks passed: 24    Warnings: 0    Checks failed: 1
Overall: FAIL
```

Figure 8 — Python checker detecting the missing VLAN 20 on SW1 Gi0/2.

The checker compares the observed allowed set 10,30,99 with the required set 10,20,30,99 and returns one mandatory failure. Its suggested fix directly identifies the required configuration change.

```
SW1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1
Gig0/2      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      10,20,30,99
Gig0/2      10,20,30,99

Port        Vlans allowed and active in management domain
Gig0/1      10,20,30,99
Gig0/2      10,20,30,99

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      10,20,30,99
Gig0/2      10,20,30,99
```

Figure 9 — Connectivity path restored after VLAN 20 was re-added to SW1 Gi0/2.

After the repair, both SW1 trunks carry 10,20,30,99 and the Guest path can again reach its gateway. Re-running the checker with the healthy capture returns the 25/0/0 PASS baseline.

F05 verification note

For F05, a successful ping to 203.0.113.10 is not accepted as proof of NAT health because INTERNET-SRV is directly connected to R1-BRANCH. The fault is verified by removing ip nat outside from G0/1, clearing translations, generating fresh Employee traffic, and confirming that no proper new translation appears. The repair restores ip nat outside, clears stale state, generates fresh traffic again, and verifies the returned translation entries.

12. PYTHON NETWORK ASSURANCE TOOL

12.1 Purpose

The Python tool provides an independent comparison between intended design and captured device state. It does not configure Packet Tracer devices; instead, it parses saved IOS outputs and reports specific mismatches in a form that supports troubleshooting and repeatable assessment.

12.2 requirements.yaml

requirements.yaml is the machine-readable source of intent. It defines the VLANs, subnets, gateways, router interfaces, NAT roles, switch management addresses, required trunks, server addresses, DHCP gateway values, Guest traffic rules, and authorized SSH source.

```
vlans:
  - id: 20
    name: GUEST
    subnet: 10.10.20.0/24
    gateway: 10.10.20.1
expected_trunks:
  - switch: SW1
    interface: Gi0/2
    allowed_vlans: [10, 20, 30, 99]
```

12.3 Input show-command files

The committed sample_inputs directory contains real captures from the completed Packet Tracer topology. Healthy and fault-specific trunk files allow the same checker to demonstrate both compliance and failure.

Input	Validation Purpose
sw1_vlan.txt / sw2_vlan.txt	Required VLAN IDs and names on both switches
sw1_trunk_good.txt / sw1_trunk_fault.txt	Healthy and F02 allowed-VLAN state on SW1
sw2_trunk.txt	Expected VLANs on SW2 Gi0/1
r1_interfaces.txt	Gateway and external interface addresses
r1_running_config.txt	NAT roles, DHCP defaults, ACL application, and VTY restriction
r1_nat.txt / r1_dhcp.txt / r1_acl.txt	Supporting operational state and policy evidence

12.4 Validation logic

checker.py validates the YAML structure, normalizes interface names, parses VLAN and trunk tables, extracts router interface addresses and configuration stanzas, evaluates NAT inside/outside roles, checks DHCP default-router values, confirms ACL 100 inbound on G0/0.20, and verifies that the VTY access-class permits only 10.10.99.10. Mandatory mismatches produce FAIL; incomplete optional evidence may produce WARN; confirmed requirements produce PASS.

Exit codes: 0 = all mandatory checks pass; 1 = at least one mandatory failure; 2 = input or configuration error.

12.5 Healthy report

Running the checker with the healthy captures confirms every mandatory requirement:

```
SmartBranch 360 Assurance Summary
==================================================
Checks passed: 25
Warnings:      0
Checks failed: 0
Overall: PASS
==================================================
```

12.6 Missing VLAN fault detection

Replacing the healthy SW1 trunk capture with sw1_trunk_fault.txt changes the result to 24 passes and one failure. As shown in Figure 8, the tool reports the expected and observed VLAN sets, explains the likely Guest symptom, and recommends restoring VLAN 20. This demonstrates that the assurance process is diagnostic rather than a simple overall status check.

13. VERSION CONTROL AND GITHUB

Git was used to maintain a structured and reproducible project history. GitHub provides a single submission location for the working topology, intended requirements, Python checker, healthy and fault reports, design documentation, and selected evidence. Local-only backups and experimental files are excluded so the public repository remains clean.

```
SmartBranch360/
├── packet_tracer/SmartBranch360.pkt
├── requirements/requirements.yaml
├── python_checker/
│   ├── checker.py
│   ├── sample_inputs/
│   └── sample_reports/
├── evidence/stage9/
├── evidence/stage10/
├── design/
└── college_submission/
```

Repository URL: deeps2710/SmartBranch360

Reproducibility

- The Packet Tracer file represents the final healthy topology.
- requirements.yaml records the intended state independently of screenshots.
- Captured IOS outputs allow the checker to be run without manual re-entry.
- Healthy and F02 fault reports demonstrate both success and targeted detection.
- Evidence folders separate normal-operation proof from fault-remediation proof.

14. DEMONSTRATION WORKFLOW

The demonstration is designed as a concise evidence-led walkthrough. It uses F02 as the single fault case so the audience can see the complete baseline-fault-detection-repair cycle without repeating all five scenarios.

Step	Demonstration Action	Evidence / Outcome
1	Introduce the Packet Tracer topology	Identify R1, SW1, SW2, VLAN zones, clients, servers, and AP-GUEST.
2	Show the healthy baseline	Employee DHCP, Employee-to-SRV1 ping, Guest external access, and Guest internal block.
3	Verify management and translation	ADMIN-PC SSH login and show ip nat translations.
4	Inject F02	Remove VLAN 20 from the SW1 Gi0/2 allowed list and reproduce Guest gateway failure.
5	Run the Python checker	Observe the specific FAIL for VLAN 20 with expected and observed sets.
6	Repair the trunk	Re-add VLAN 20, verify show interfaces trunk, and repeat the Guest test.
7	Close with the healthy report	Re-run the checker and show 25 PASS, 0 WARN, 0 FAIL.

Demo video folder: Open the SmartBranch 360 demonstration folder

The repository remains the authoritative source for the full evidence set, while the video provides a guided demonstration of the most important behaviours.

15. CHALLENGES AND LEARNING OUTCOMES

15.1 Challenges

- Maintaining consistent VLAN creation and allowed lists across two switches and three required trunk interfaces.
- Coordinating router subinterfaces, DHCP gateways, DNS assignments, NAT roles, and ACL placement so that services worked together rather than in isolation.
- Preserving Guest external access while denying access to Employee, Server, and Management networks.
- Distinguishing application-layer DNS failure from basic IP reachability during F04.
- Using translation evidence, rather than ping alone, to verify the revised F05 NAT Outside Missing scenario in a directly connected simulation.
- Parsing variable Cisco IOS command formats robustly enough for repeatable automated checks.
- Selecting a small set of screenshots that proves the major requirements without making the report visually repetitive.

15.2 Learning Outcomes

- Designed and implemented role-based VLAN segmentation and IPv4 subnetting for a branch office.
- Configured access ports, 802.1Q trunks, router-on-a-stick, DHCP, DNS, guest wireless access, NAT/PAT, ACLs, and SSH.
- Applied positive and negative testing to distinguish expected security denials from network faults.
- Used Cisco IOS operational commands to isolate faults at the endpoint, Layer 2, Layer 3, service, and translation layers.
- Represented network intent in YAML and validated captured state with Python and PyYAML.
- Produced repeatable technical evidence, reports, and a structured GitHub repository suitable for academic and portfolio review.

16. CONCLUSION

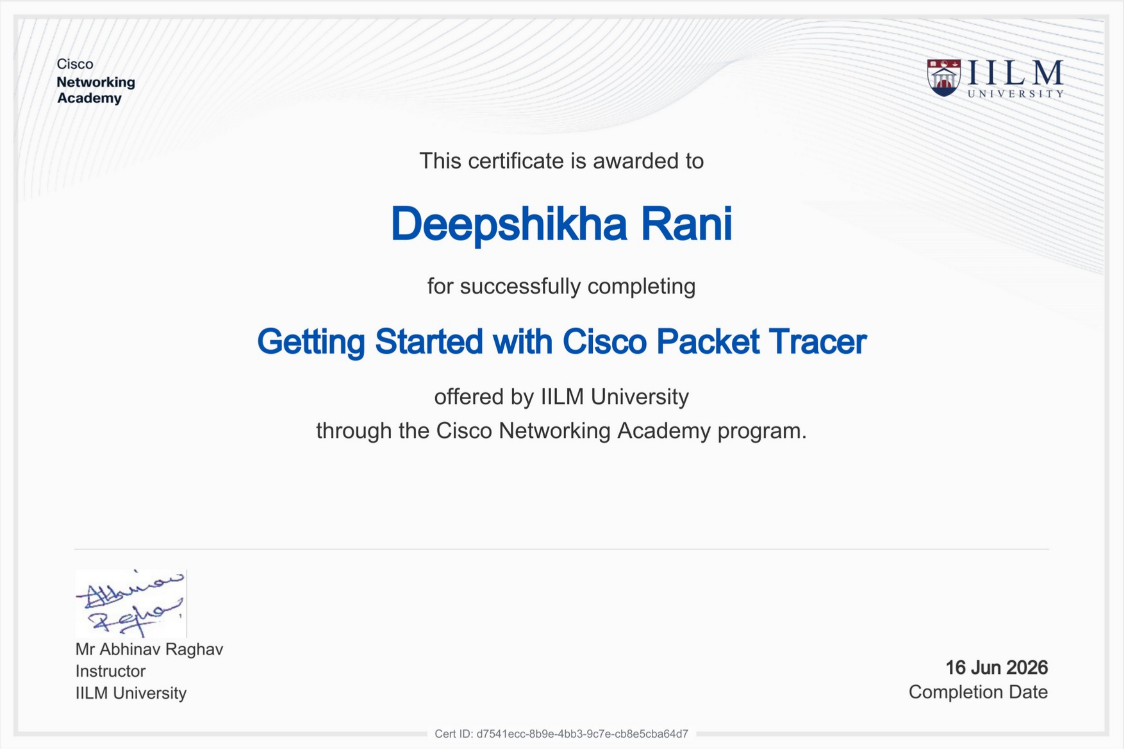
SmartBranch 360 demonstrates the complete lifecycle of a secure small-branch network: planning, segmentation, routing, addressing, service delivery, wireless access, external translation, policy enforcement, secure management, validation, troubleshooting, and configuration assurance. The final Packet Tracer topology meets the intended Employee, Guest, Server, and Management requirements while keeping each trust zone explicit.

The testing results show that required traffic succeeds and unauthorized Guest or management traffic is blocked. The five fault scenarios demonstrate that different configuration errors create distinguishable symptoms and can be repaired through evidence-based troubleshooting. The revised NAT fault further shows why the verification method must match the topology rather than relying on a convenient but inconclusive ping test.

Finally, the Python assurance tool strengthens the project by turning design intent into repeatable checks. A healthy run records 25 passes with no warnings or failures, while the F02 capture produces a precise failure for the missing Guest VLAN. Together, the topology, IOS evidence, YAML requirements, automation, Git history, demonstration video, and certificates form a complete professional internship submission.

Final project status: Healthy baseline verified; all five controlled faults repaired; assurance result 25 PASS / 0 WARN / 0 FAIL.

INTERNSHIP CERTIFICATES



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This certificate is awarded to

Deepshikha Rani

for successfully completing

Getting Started with Cisco Packet Tracer

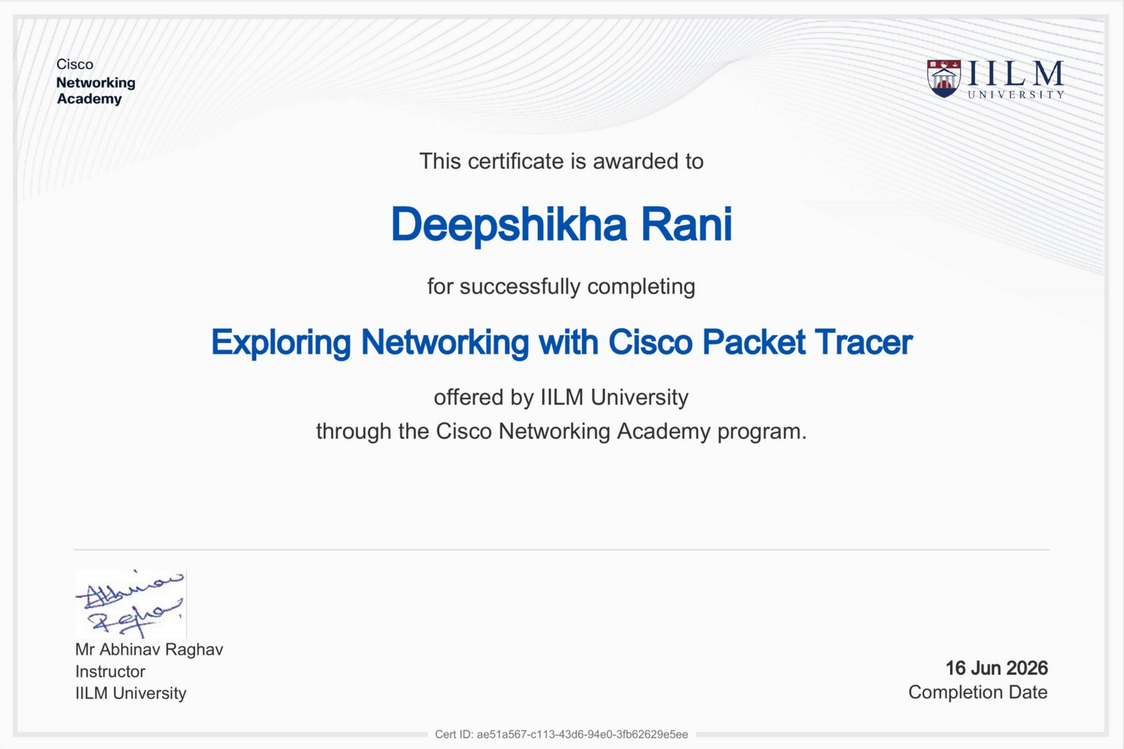
offered by IILM University
through the Cisco Networking Academy program.

Mr Abhinav Raghav
Instructor
IILM University

16 Jun 2026
Completion Date

Cert ID: d7541ecc-8b9e-4bb3-9c7e-cb8e5cba64d7

Certificate 1 — Getting Started with Cisco Packet Tracer; completed 16 June 2026.



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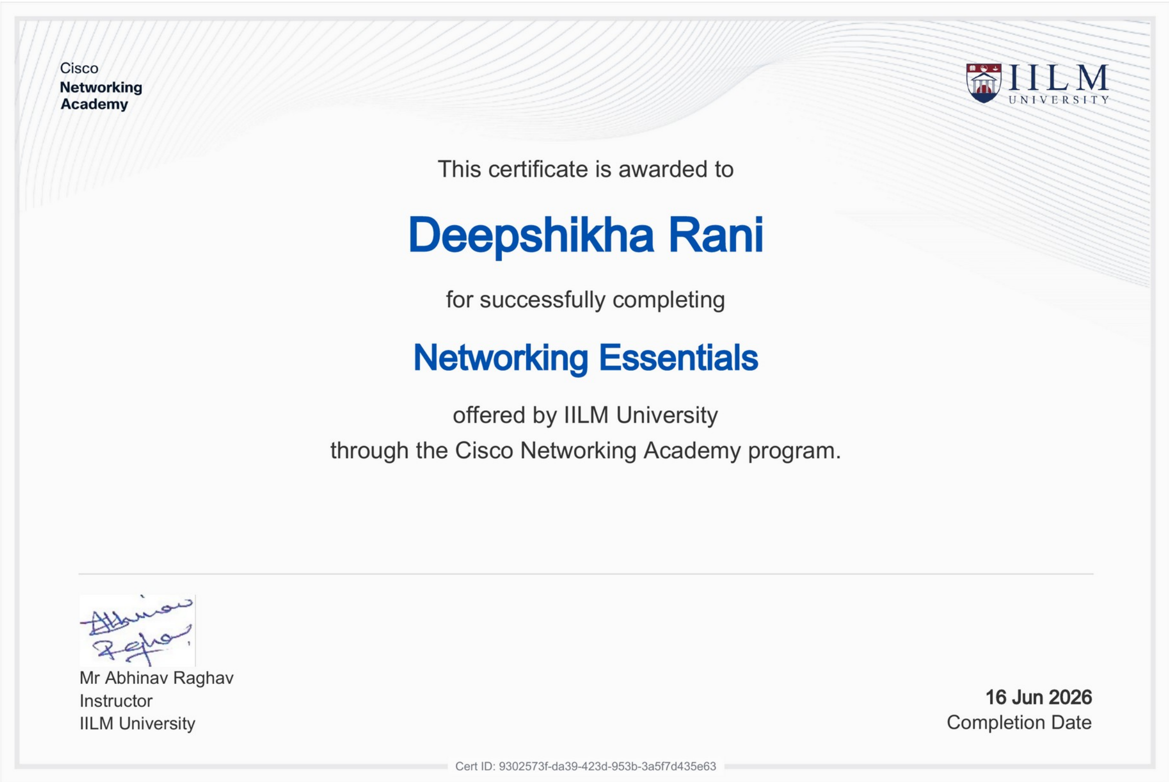
INTERNSHIP CERTIFICATES (CONTINUED)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